

## **LAB No. 6**

**Aim:** Create an Art with Neural style transfer on a given image using deep learning.

### **Objective:**

The objective of this lab assignment is to apply neural style transfer using deep learning techniques to create artistic images by blending the content of a given image with the style of another image.

### **Materials Required:**

1. Computer with Python and TensorFlow installed
2. Jupyter Notebook or any Python IDE
3. Required Python libraries: TensorFlow, NumPy, Matplotlib
4. Given content image
5. Style image(s) for artistic transformation

### **Procedure:**

1. Introduction to Neural Style Transfer:
  - Explain the concept of neural style transfer, which involves generating an image that combines the content of one image with the style of another image.
  - Briefly discuss the deep learning techniques involved, such as convolutional neural networks (CNNs) and optimization algorithms.
2. Setting up the Environment:
  - Ensure that Python and TensorFlow are installed on the computer. - Open a Jupyter Notebook or any Python IDE for coding.
3. Loading the Images:
  - Load the given content image and the style image(s) into the program using suitable libraries such as TensorFlow or NumPy.
4. Understanding the Pre-trained Model:
  - Introduce the pre-trained VGG19 model, which is commonly used for neural style transfer due to its effectiveness in capturing image features.

#### 5. Defining the Loss Functions:

- Explain the concept of content loss and style loss.
- Implement functions to calculate the content loss and style loss based on the features extracted from the pre-trained VGG19 model.

#### 6. Creating the Total Loss:

- Combine the content loss and style loss to create the total loss, which will be minimized during the optimization process.

#### 7. Optimization Process:

- Utilize an optimization algorithm (e.g., Adam optimizer) to minimize the total loss and generate the final artistic image.
- Iterate through multiple iterations to refine the generated image and enhance its artistic qualities.

#### 8. Displaying the Results:

- Visualize the generated artistic image at regular intervals during the optimization process to observe its transformation.
- Display the final artistic image alongside the original content image and the style image for comparison.

#### 9. Experimentation and Analysis:

- Encourage students to experiment with different content and style images, as well as adjust parameters such as learning rate and number of iterations, to observe variations in the generated artistic images.
- Prompt students to analyze the results and discuss the impact of different factors on the quality and style of the generated images.
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### **Conclusion:**

In this lab assignment, students learned how to apply neural style transfer using deep learning techniques to create artistic images. By blending the content of a given image with the style of another image, they gained insights into the capabilities of convolutional neural networks and optimization algorithms in generating visually appealing artwork.